



National University of Computer & Emerging Sciences, Karachi

Fall-2023 Department of Computer Science

Final Exam

26th December 2023, 09:00 AM – 09:30 AM

Part (A)



Course Code: CS2009	Course Name: Design and Analysis of Algorithm
Instructor Name / Names: Dr. Muhammad Atif Tahir, Dr. Fahad Sherwani, Dr. Farrukh Salim, Dr. Nasir Uddin, Mr. Faisal Ahmed, Mr. Sandesh Kumar, Ms. Ansum Hamid, Mr. Syed Faisal Ali, and Mr. Minhal Raza	
Student Roll No:	Section:

Instructions:

- Must be submitted within 30 minutes. No extra sheets allowed. Must solve in given space
- You can start Part B but not before returning Part A to invigilator.

Time: 30 minutes

Max Marks: 10

Question # 1 [CLO: 2]

[5 marks]

Problem	Give name of Algorithm e.g. insertion sort etc [0.25 marks each]	Time [0.125 marks]	Space [0.125 marks]
to design networks with the least cost. It can be used to find the least expensive network connections that can connect all the nodes in the network.			
selecting sites for satellite stations in order to maximize the global traffic with respect to a budget constraint.			
to avoid vehicle collisions in cars and airplanes.			
Arbitrage detection in financial markets, where negative weights represent profitable trading opportunities.			
to design a scheduler that to schedule a set of tasks. A number of the tasks need to wait for some other tasks to complete prior to running themselves.			
Picking the optimal location for a cell tower so that it covers the maximum number of customers			
Consider a social network, where the graph's vertices represent people, and the graph's edges represent mutual acquaintance.			
scheduling of a machine to drill holes in a circuit board or other object. In this case the holes to be drilled are the cities, and the cost of travel is the time it takes to move the drill head from one hole to the next.			

sorts data with integer keys by grouping the keys by individual digits that share the same significant position and value (place value)			
sorts data by selecting a 'pivot' element from the array and partitioning the other elements into two sub-arrays, according to whether they are less than or greater than the pivot			

Question # 2 [CLO: 2]

[5 marks]

(a) [3 Points] For each of the following questions, indicate whether it is T (True) or F (False) and justify using some examples e.g. assuming a function?

I. For all positive $f(n)$, $g(n)$ and $h(n)$, if $f(n) = O(h(n))$ and $f(n) = \Omega(g(n))$, then $g(n) + h(n) = O(f(n))$.

II. Let $f(n)$ and $g(n)$ be asymptotically nonnegative functions, then $\min(f(n), g(n)) = \Theta(f(n) + g(n))$.

III. if $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$, then we have $(f(n))^3 = (g(n))^3$

(b) [2 Points] Compute the time complexity of the following. Show all steps clearly

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for i=1...n do
  j=n
  while i<j do
    if j mod 2 = 0 then j=j-1
    else j=i

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